Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **production AI systems**, **asynchronous inference workflows**, developer-facing APIs, and full-stack applications using Python, React, Docker, cloud infrastructure, and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Restructured a **fault-tolerant asynchronous LLM orchestration** system with explicit job states and failure isolation, processing 1K+ inference tasks while tripling throughput under concurrent production workloads and preventing cascading failures.
- Built FastAPI data integrations across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with incremental loads, reconciliation, error handling, and CI/CD-backed releases at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Engineered a **multimodal inference platform** integrating GPT-4o, Claude, Qwen-VL, and OCR to evaluate 10K+ text and handwritten submissions daily with 90%+ grading consistency across production workflows reliably.
- Designed concurrent batch processing with multithreading, secure file ingestion, AWS RDS, and REST APIs, reducing feedback turnaround 60% while sustaining stable responses during peak workloads for 150+ instructors.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, California*

- Built a PyTorch simulation pipeline processing 100K+ trajectory logs, improving long-horizon planning 18% through reproducible feature transformations, benchmark automation, regression testing, and detailed behavioral failure analysis workflows.
- Developed a spatial-temporal encoder combining map geometry, historical trajectories, and agent context, improving closed-loop stability 15% across complex scenarios through controlled evaluation and systematic root-cause debugging workflows.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Engineered cross-platform microservices with automated VM builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments reliably.
- Optimized object-oriented API integrations and AWS data flows, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20% through reusable services and structured reviews.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight microservices spanning authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across distributed production workflows.
- Built responsive React workflows and automated Docker deployments through Jenkins CI/CD, achieving 95%+ successful validation and cutting release time 50% through repeatable configurations, staged testing, and monitoring.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Bash
AI & Backend: PyTorch, Transformers, LLMs, FastAPI, Flask, Node.js, REST APIs, WebSockets, Async Processing, Microservices
Frontend & Data: React, Next.js, PostgreSQL, SQL Server, MySQL, MongoDB, Redis, Vector Search, FAISS
Infrastructure & Quality: Docker, Kubernetes, AWS, Azure, GCP, Linux, Git, Jenkins, CI/CD, Unit Testing, Integration Testing, Debugging, Monitoring, Performance Optimization

Projects

CaptionCraft - Real-Time ML Application Live | *PyTorch, Transformers, Flask*

- Benchmarked BLIP, GIT, and ViT-GPT-2 across 10K images and 100K captions, improving measured accuracy from 81% to 96% while building scalable Flask inference APIs for responsive generation.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*

- Built a semantic retrieval system over 50K+ documents using transformer embeddings and FAISS, enabling contextual generation with sub-second search through optimized indexing, similarity pipelines, and metadata persistence.

Linguistix | *Python, AWS ECS/Fargate, Docker, CI/CD*

- Developed and deployed a containerized RESTful data-processing API on AWS ECS/Fargate with CodePipeline releases, automated validation checks, 87% measured accuracy, and reliable repeatable production deployment workflows across environments.